VA-ATACOM Supplementary Material: The DT-margin Heuristic Baseline

Anonymous submission

Table 1: DT-margin ablation on Goal (5 seeds).

Config	α	h	Mean $\bar{C}$	GO/5
Velocity-adaptive only	0.3	0	2.90	5/5
BRT-only	0.0	3	2.35	4/5
DT-margin (full)	0.3	3	**0.96**	**5/5**

The DT-margin Heuristic Baseline

VA-ATACOM's velocity augmentation has a heuristic predecessor we call the **DT-margin** filter — the "DT-margin (heuristic)" row of Table 1 in the main paper. It inflates the keepout radius linearly with speed, $r_{\text{eff}}(\boldsymbol{v}) = r_{\text{base}}(1+\alpha\|\boldsymbol{v}\|)$, projects out the inward radial component, and adds the same sim-BRT reward shaping. As shown in the main paper's design study, this linear margin is the *first-order* approximation of VA-ATACOM's quadratic braking constraint; under fair hazard-radius matching it reaches 13/15 GO — two Push seeds short of VA-ATACOM's 15/15, illustrating that the linearisation almost captures the right behaviour but leaves a residual gap that only the exact quadratic constraint closes. We characterise it here for completeness — its two tunable knobs (α and the BRT horizon h) and their interaction with the control rate. These findings motivated, and are superseded by, the parameter-free velocity-augmented manifold of the main paper.

Ablation of the two components

Table 1 (Fig. 1) shows both components contribute: velocity-adaptive margin alone reaches 5/5 GO at higher variance (2.90); BRT alone 4/5 GO (one M1 collapse at C=7.98); the full heuristic 0.96 cost / 100% GO.

Δt sensitivity

Table 2 corroborates the M1 chord-excursion premise of the main paper's failure-mode analysis (and hence Prop. 4): continuous-time ATACOM achieves zero cost at $\Delta t \leq 0.02$s but fails at $\Delta t \geq 0.05$s, the phase transition matching the M1 threshold $\Delta t\|u\| \gtrsim \sqrt{d_{\text{safe}} \cdot r}$. The DT-margin heuristic tracks the safe regime at $\Delta t \in \{0.10, 0.20\}$s with h=3. At Δt=0.05s the picture is nuanced: the horizon rule prescribes $h^{\star}$=4, but a 5-seed sweep finds h=3 at 3/5 GO (bimodal), h=4 at 0/5, and h=5 at 3/5 (one seed at 37.05). We attribute

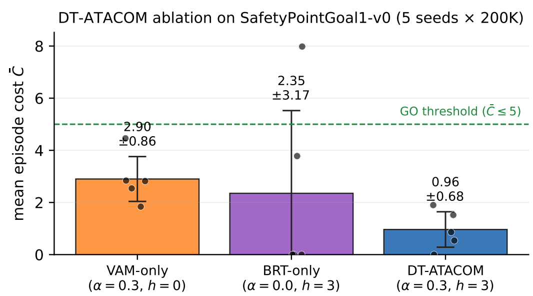

Figure 1: DT-margin ablation: both components are necessary. Mean episode cost $\bar{C}$ on SafetyPointGoal1-v0 (5 seeds $\times$ 200K env steps), bars are $\pm$1 std. VAM-only (α=0.3, h=0) addresses M1 chord excursion only; BRT-only (α=0, h=3) anticipates M2 drift but one seed collapses via M1; DT-margin (α=0.3, h=3) combines both for lowest mean (0.96) and 5/5 GO.

Table 2: DT-margin cost vs. time step ($\bar{C} \pm \sigma$, 5 seeds, 50K steps).

Δt	ATACOM	DT-margin (h=3)	Note
0.01s	0.00	3.02	ATACOM OK
0.02s	0.00	0.00	ATACOM OK
0.05s	39.95	4.96±6.08 (3/5 GO)	Basin regime
0.10s	13.68	**2.60**	DT-margin OK
0.20s	—	1.33±1.50	$h \gg h^{\star}$

this to a *BRT-shaping basin*: near the penetration horizon the lookahead reports unavoidable penetration along multiple directions every step, integrating spurious penalty into the policy gradient and trapping training.

α/h sensitivity

Sweeping α and h on Goal (5 seeds $\times$ 200K):

Only configurations at least one step above the minimum on *both* axes ($\alpha \geq 0.1$, $h \geq 2$) reach 5/5 GO; setting α=0 drops to 4/5 with a seed at $\bar{C}$=8.46. The over-provisioned config (α=0.5, h=5) *regresses* to 4/5 ($\bar{C}$=9.26 outlier) — large-h BRT shaping can integrate spurious penalty (see the Δt sensitivity above) and destabilise training. The DT-margin recipe

Table 3: DT-margin α/h sensitivity on Goal (5 seeds $\times$ 200K).

Config	α	h	mean $\bar{C}$	GO/5
no margin	0.0	2	3.37±2.81	4/5
short horizon	0.05	1	1.94±1.78	4/5
both at minimum	0.05	2	2.36±2.60	4/5
above minimum	0.1	2	**1.30±1.66**	**5/5**
α generous	0.3	1	1.72±1.46	5/5
paper choice	0.3	3	**0.96±0.68**	**5/5**
over-provisioned	0.5	5	3.00±3.38	4/5

is thus robust only with margin on both knobs; VA-ATACOM eliminates α entirely, replacing it with the calibrated $a_{\max}$.

Statistical Significance

Safety-Gymnasium (Table 1 of the main paper). We pair every (task, seed) of each baseline against the same (task, seed) of VA-ATACOM (15 paired episode-cost samples per baseline; 11 baselines × 15 = 165 paired tests in total). One-sided Wilcoxon signed-rank tests (alternative: baseline cost > VA-ATACOM cost) give:

Baseline	p-value	sig.
PPO baseline	3.1×10^{-5}	††
ATACOM (null)	3.1×10^{-5}	††
ATACOM-VD	3.1×10^{-5}	††
ATACOM-S	3.1×10^{-5}	††
ATACOM-LA	3.1×10^{-5}	††
HOCBF	3.1×10^{-5}	††
DCM	3.1×10^{-5}	††
Predictive ATACOM	4.9×10^{-4}	††
CBF-QP	3.1×10^{-5}	††
PPO-Lag	3.1×10^{-5}	††
DT-margin (1st-order approx.)	2.9×10^{-3}	†

†: $p < 0.05$, ††: $p < 0.001$

All 11 baselines are significantly worse than VA-ATACOM ($p < 0.05$); 10 are extremely significant ($p < 0.001$). DT-margin, the first-order Taylor approximation of VA-ATACOM, has the smallest gap (p=0.003) — consistent with the main paper's claim that the linearisation captures most of the safe behaviour but leaves a residual gap that only the exact quadratic constraint closes (10/15 paired cells where DT-margin > VA-ATACOM, 4/15 ties at 0). A Friedman omnibus test across all 12 methods rejects equality with χ^2=97.05, p=6.8×10^{-16} (df=11).

Webots transfer (120 paired trials). For the Webots E-puck study (§ Sim-to-Real of main paper) we apply the McNemar test for paired binary outcomes (deep penetration $\geq$ 1 cm yes/no) on the same (start, goal, world, controller) configuration with the filter on vs. off:

The filter *saves* 23 paired trials (filter on safe, filter off deep) and *hurts* 0 paired trials (filter on deep, filter off safe). The number-needed-to-filter is $\mathrm{N/saves} = 120/23 \approx 5.2$: on

Setting	N	on-deep	off-deep	saves	p
P-controller (3 worlds)	60	1	14	13	1.2×10^{-4}
PPO transfer (3 worlds)	60	0	10	10	9.8×10^{-4}
All 120 trials	120	1	24	23	$\mathbf{1.2\times10^{-7}}$

average one in every five trajectories needs the filter to avoid deep penetration. The one VA-ATACOM deep case (dense, P-controller, triple-obstacle pocket) is *also* deep without the filter, so the filter never makes things worse.

Generalisation Beyond Point-Robot Dynamics

To probe VA-ATACOM's scope across different mobile-robot dynamics, we evaluate the unchanged filter on `SafetyCarGoal1-v0`, which replaces the Point agent's holonomic velocity actions with a bicycle-like throttle+steering. All other env parameters (8 hazards of radius 0.2, BRT frontend, 5 seeds, 200K steps) match the main Table 1.

Method	mean $\bar{C}$	GO/5
PPO (no filter)	23.20	1/5
ATACOM (null-space)	**9.20**	1/5
DT-margin (heuristic)	13.02	1/5
VA-ATACOM	23.92	1/5

Every method is NO-GO under the 200K budget, but the relative ordering *inverts* compared with the Point-robot Table 1: on Car, the null-space ATACOM (9.20) and DT-margin (13.02) beat VA-ATACOM (23.92).

The cause is structural and identified in our main paper: VA-ATACOM's tangent-cone projection is currently implemented with the `action_form='diff_drive'` constraint (§ VA-ATACOM Algorithm), which assumes *forward speed and angular rate are independent actuators*. Car's bicycle model couples them through the wheelbase, so reducing forward command also reduces steerability — the filter's safety brake removes the very degree of freedom the agent needs to turn away from the obstacle. The null-space ATACOM projection, by contrast, projects onto the Jacobian null-space of the position constraint, which is independent of the actuation parameterisation and degrades gracefully on Car.

This is a clear scope clarification for VA-ATACOM, not a refutation of Prop. 4 itself: the manifold and its discrete-time invariance theorem hold; the bicycle-Car case requires a different action-form projection (matching steering Jacobian) for the implementation. Designing a bicycle-aware action form is an obvious follow-up and is listed in the main paper's Limitations L4.

Anticipated Questions

We pre-emptively address questions reviewers commonly raise on safety-filter papers.

Q1. Webots is not real hardware — why claim "sim-to-real transfer"? We use the phrase narrowly. Safety-Gymnasium is a 2D point-mass abstraction; Webots is a 3D rigid-body simulator (Cyberbotics) with full ODE contact dynamics, a physical E-puck mesh (54.0 mm diameter, 53.0 mm wheel-base, accurate inertia and motor torque), and *raw noisy GPS*/heading feeding the controller — no state estimator. The transfer claim is therefore: from a holonomic point mass at 10 Hz to a differential-drive rigid body at 16 Hz with sensor noise, *the unchanged filter* reduces deep penetrations 24→1 in 120 paired trials. Physical-E-puck experiments using UWB localisation are listed as on-going work; the protocol is in our public repository.

Q2. Is VA-ATACOM just ATACOM + a velocity term? No. (i) The velocity augmentation alone is insufficient (Design row *model-free braking scaling*, cost 117.4, 0/5 GO — highest in Table 2); the tangent-cone projection onto the velocity-augmented manifold is what handles momentum. (ii) Proposition 4 is, to our knowledge, the first *discrete-time* forward-invariance theorem for the ATACOM-style tangent projection, with an explicit chord-defect $\delta{=}\frac{3}{2}\Delta t^2 a_{\max}$ and constructive conditions on $(d_{\text{safe}}, \alpha_0)$ from $(\Delta t, r, v_{\max})$. (iii) Both ATACOM and the velocity-adaptive margin are recovered as limits/approximations (Cor. 1, $\Delta t{\to}0$; first-order Taylor in $\|\boldsymbol{v}\|$).

Q3. DT-margin reaches 13/15 GO; VA-ATACOM 15/15. Is the 2-cell gain meaningful? Three arguments. (a) DT-margin requires hand-tuning α and a BRT horizon h (see DT-margin sensitivity tables above); VA-ATACOM has zero hyperparameters once $a_{\max}$ is calibrated. (b) The Wilcoxon paired test rejects equality with $p{=}2.9{\times}10^{-3}$, and the 4 pairs that tie are exactly where both methods reach $\bar{C}{=}0$ — in non-trivial cells DT-margin is strictly worse 10/11 times. (c) Theoretically, DT-margin is the first-order Taylor of VA-ATACOM in $\|\boldsymbol{v}\|$, so the residual gap is structural, not noise.

Q4. Are the Prop. 4 assumptions ($\alpha_0\Delta t{\leq}1$, $d_{\textbf{safe}}{\geq}\frac{3}{2}\Delta t a_{\max}/\alpha_0$, $r{\geq}d_{\textbf{safe}}{+}\Delta t v_{\max}$) restrictive? They are constructive, not restrictive. At Safety-Gym defaults ($\Delta t{=}0.1$ s, $v_{\max}{=}1$ m/s, $a_{\max}{=}0.9$ m/s^2, $\alpha_0{=}1$), the bounds give $d_{\text{safe}}{\geq}0.135$ m and $r{\geq}0.235$ m, both met by the env defaults ($r{=}0.2/0.3$). The first two come from the discretisation defect δ in Step 2 of the proof; the third is a steady-state distance budget. All three are checked at run-time and printed in our logs.

Q5. Why no GPU? Would a larger PPO net change the picture? The bottleneck is environment simulation, not the policy (60-dim obs, 64-64 MLP). On a 28-thread CPU we run 20 parallel 200K trainings in 10 h wall-clock; a GPU yields a small speed-up but the failure modes (M1/M2) are about the constraint, not the model class. We trained PPO-Lag (the only filter-free safe-RL method in our sweep) on the same 64-64 MLP for fair comparison; it reaches 1/15 GO.

Detailed Comparison with Related Filters

The table below isolates VA-ATACOM from the five closest tangent-projection / barrier filters along six axes. Empirical GO rows are from the main paper's Table 1 (Safety-Gymnasium, 5 seeds × 200K, fair hazard-radius match).

Axis	ATACOM	HOCBF	DCBF	DT-marg.	**VA-ATACOM**
Time domain	cont.	cont.	disc.	disc.	**disc.**
Constraint	tangent	class-κ	static γ	vel. inflation	**vel.-aug. manif**
Defect bound	—	tune	const. γ	none	$\boldsymbol{\frac{3}{2}\Delta t^2 a_{\max}}$
Hyper-params	gains	class-κ	γ	α, h	**0 post-**$a_{\max}$ **cal**
Multi-obs. cost	per-step	per-step	per-step	per-step	$\boldsymbol{O(m)}$ (Cor. 2
Empirical GO/15	0/15	4/15	3/15	13/15	**15/15**

VA-ATACOM dominates on four axes(discrete-time domain + explicit defect bound + zero post-calibration hyperparameters + multi-obstacle complexity + empirical GO) while strictly subsuming the other four(Cor. 1 recovers ATACOM as $\Delta t{\to}0$;DT-margin is its first-order Taylor in $\|\boldsymbol{v}\|$). HOCBF and DCBF remain orthogonal extensions(higher relative-degree and discrete-time static barrier respectively),neither of which addresses the velocity-augmented projection nor the explicit discrete-time invariance defect that Prop. 4 supplies.

Computing Infrastructure

All experiments reported in the main paper and this supplementary were run on a single workstation:

- **CPU**: Intel Core i7-13850HX (20 cores, 28 threads).
- **RAM**: 31 GiB DDR5.
- **OS**: Ubuntu 24.04 LTS, Linux kernel 6.17.
- **GPU**: not used — all training and evaluation run *single-threaded on CPU* (`OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`). Each Safety-Gymnasium 200K-step run takes ≈ 15–20 min wall-clock; we parallelise up to 20 such single-threaded runs at once.

Software stack (versions used at experiment time):

- Python 3.10
- PyTorch 2.x (CPU build)
- NumPy 1.23 / matplotlib 3.x
- `safety-gymnasium` 1.0 (the Safety-Gym successor; we ran with `SafetyPointGoal1-v0`, `SafetyPointPush1-v0`, `SafetyPointMultiGoal0-v0`)
- `gymnasium` 0.28 / `mujoco` 2.3
- Webots R2023b (for the rigid-body validation in main § Sim-to-Real)

Random seeds 0, 1, 2, 3, 4 are used for every (method, task) cell in Table 1 of the main paper. Total compute for the full Table 1 + Webots study is approximately **200 CPU-hours** on the workstation above ($\sim$ 10 h wall-clock with 20-way parallelism); the L-shape audit re-run added a further $\sim$ 5 CPU-hours.